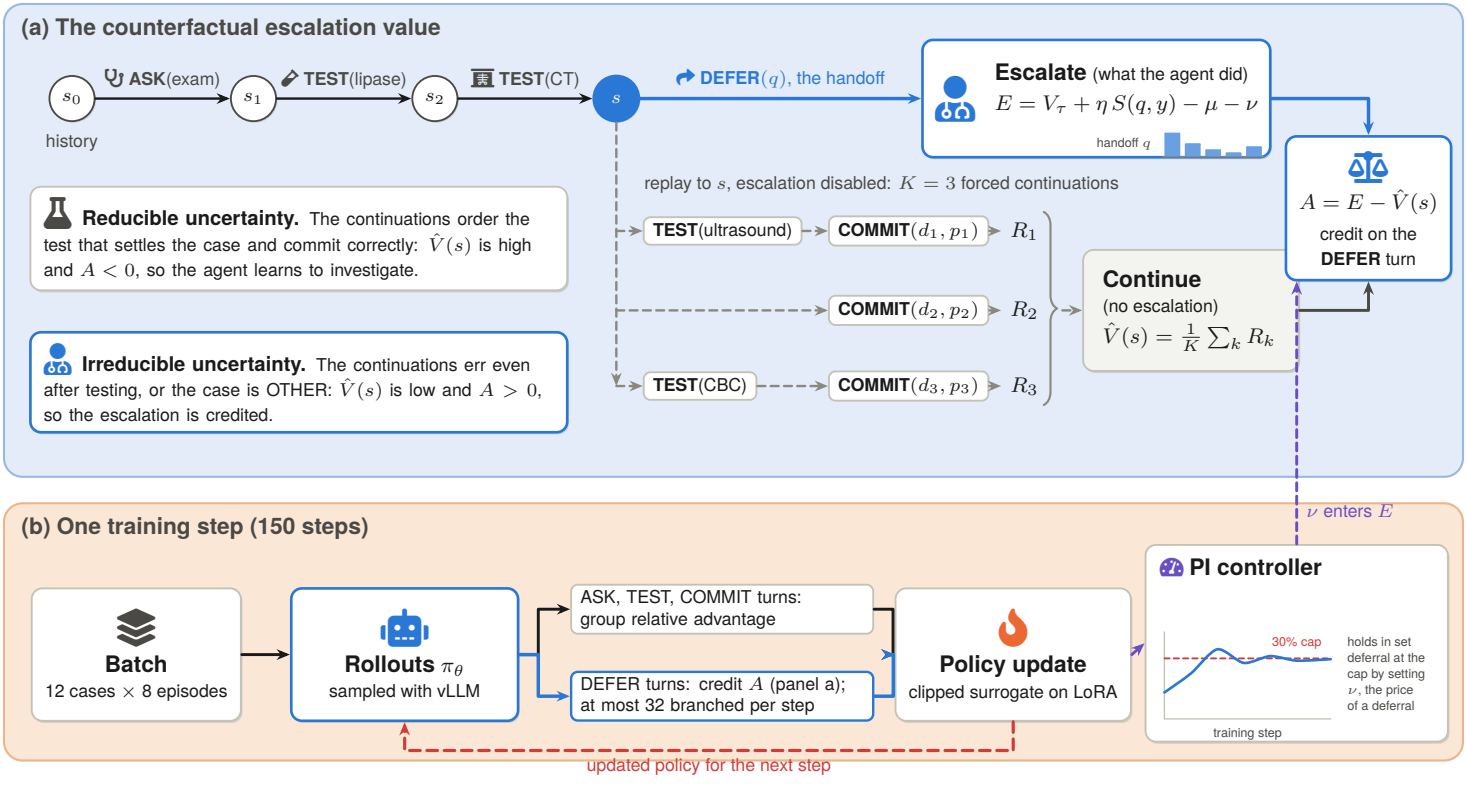
(a) The counterfactual escalation value
s_0
ASK(exam)
s_1
TEST(lipase)
s_2
TEST(CT)
s
history
DEFER(q), the handoff
Escalate (what the agent did)
$E = V_\tau + \eta\, S(q,y) - \mu - \nu$
handoff q
$A = E - \hat{V}(s)$
credit on the DEFER turn
Reducible uncertainty. The continuations order the test that settles the case and commit correctly: $\hat{V}(s)$ is high and $A < 0$, so the agent learns to investigate.
Irreducible uncertainty. The continuations err even after testing, or the case is OTHER: $\hat{V}(s)$ is low and $A > 0$, so the escalation is credited.
replay to s, escalation disabled: $K = 3$ forced continuations
TEST(ultrasound)
COMMIT(d_1, p_1)
R_1
COMMIT(d_2, p_2)
R_2
TEST(CBC)
COMMIT(d_3, p_3)
R_3
Continue
(no escalation)
$\hat{V}(s) = \frac{1}{K}\sum_k R_k$
(b) One training step (150 steps)
ν enters E
PI controller
30% cap
holds in set deferral at the cap by setting ν, the price of a deferral
training step
Batch
12 cases × 8 episodes
Rollouts π_θ
sampled with vLLM
ASK, TEST, COMMIT turns: group relative advantage
DEFER turns: credit A (panel a); at most 32 branched per step
Policy update
clipped surrogate on LoRA
updated policy for the next step